A 48-qubit code with native CCZ

Boolean functions, projective geometry, and SAT

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Why a native CCZ gate

Fault-tolerant computation needs a non-Clifford logical gate. CCZ changes the sign of $|111\rangle$ only, and Clifford gates plus CCZ are universal.

Magic-state distillation

Standard route. Prepares noisy resource states and distills them, at a large space-time overhead.

Transversal gate

A code whose physical phase rotations implement the logical gate directly on the data block.

Target: a CSS code with $k = 3$ logical qubits and distance $d \geq 3$ whose logical CCZ is one layer of physical T or $T^\dagger$, one gate per data qubit, with no ancilla and no correction. We call such an implementation *native*. A single-qubit gate cannot spread one fault to two qubits.

Question: how short can a native CCZ code with $d \geq 3$ be?

Bravyi–Haah 2012 (distillation) Campbell–Howard 2017 (CSS codes with transversal CCZ).

Results: known and open lengths

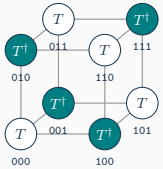
Here n is the number of physical qubits, $k = 3$, and $d_Z \geq 3$.

length	what is known
$n \leq 38$	no CSS code meets the gate conditions (this work, a lower bound)
$39 \leq n \leq 46$	open
$n = 47$	Jacinto et al. code meets the conditions with a Clifford correction, but has no native pattern (this work)
$n = 48$	the code Q_{48}: native CCZ by $26\,T + 22\,T^\dagger$ (this work)

Method: projective geometry chooses the stabilizers, a SAT solver finds the logical operators, and a linear system over $\mathbb{Z}_4$ finds the $T/T^\dagger$ signs.

Q_{48} is the shortest known native code. The lower bound does not prove 48 minimal.

Warm-up: the eight-qubit cube code



Qubits at addresses $u \in \mathbb{F}_2^3$. White: $p(u) = 0$, apply T. Teal: $p(u) = 1$, apply $T^\dagger$.

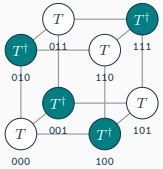
For $a \in \mathbb{F}_2^3$ let $f_a(u) = a_1u_1 \oplus a_2u_2 \oplus a_3u_3$.

$$|a\rangle_L = \tfrac{1}{\sqrt{2}}\big(|f_a\rangle + |f_a \oplus \texttt{11111111}\rangle\big).$$

Pattern: $p(u) = u_1 \oplus u_2 \oplus u_3$. With $\omega = e^{i\pi/4}$, a word v gets the phase $\omega^{\mathrm{sw}(v)}$,

$$\mathrm{sw}(v) = \#\{\text{ones at } T\} - \#\{\text{ones at } T^\dagger\} = |p \oplus v| - 4.$$

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$$\mathrm{sw}(v) = \#\{\text{ones at } T\} - \#\{\text{ones at } T^\dagger\} = |p \oplus v| - 4.$$

Per site, $(p \oplus v) - p$ equals v where $p = 0$ and $-v$ where $p = 1$. Every nonconstant affine function of three bits has exactly four ones.

label	$p \oplus f_a$, $p \oplus f_a \oplus 1$	sw	phase
$a \neq 111$	nonconstant affine	$0,\ 0$	$+1$
$a = 111$	00000000, 11111111	$-4,\ +4$	$\omega^{\pm 4} = -1$

The cube code has distance two

Write Z_u for a phase flip on qubit u. On a word v, $Z_{000}Z_{100}$ gives the sign $(-1)^{v(000)\oplus v(100)}$. Both words of $|a\rangle_L$ give a_1, so

$$Z_{000}Z_{100}|a\rangle_L = (-1)^{a_1}|a\rangle_L.$$

The only X-type stabilizer is $X^{\otimes 8}$. It commutes with this weight-2 error, so the error is undetected and acts as a logical Z_1.

$$[\![8, 3, 2]\!], \qquad d_Z = 2, \quad d_X = 4.$$

Goal: keep the gate mechanism and raise d_Z to 3. That requires a larger code and a systematic criterion.

A CSS code as a binary matrix

$$G = \begin{bmatrix} K \\ S \end{bmatrix}, \qquad \mathcal{C}_2 = \operatorname{rowspan} S \subset \mathcal{C}_1 = \operatorname{rowspan} G.$$

The s rows of S are X-stabilizer generators, the 3 rows of K are X-logical generators, and the n columns are the physical qubits. The logical basis state with label $x \in \mathbb{F}_2^3$ is a coset sum:

$$|x_L\rangle = 2^{-s/2} \sum_{y \in \mathbb{F}_2^s} |K^\top x \oplus S^\top y\rangle.$$

Distances: $d_X = \min\{|c| : c \in \mathcal{C}_1 \setminus \mathcal{C}_2\}$ and $d_Z = \min\{|c| : c \in \mathcal{C}_2^\perp \setminus \mathcal{C}_1^\perp\}$, with $|c|$ the Hamming weight.

Cube code: $S = (11111111)$, and K has rows $u_1 = 00001111$, $u_2 = 00110011$, $u_3 = 01010101$.

The logical gate becomes a phase condition

A pattern $\Gamma \in \{\pm 1\}^n$ applies T where $\Gamma_j = +1$ and $T^\dagger$ where $\Gamma_j = -1$. A basis word c gets the phase $\omega^{\mathrm{sw}_\Gamma(c)}$, with

$$\omega = e^{i\pi/4}, \qquad \mathrm{sw}_\Gamma(c) = \sum_j \Gamma_j c_j \quad \text{(signed weight)}.$$

Native CCZ condition: for every logical label x and stabilizer label y,

$$\mathrm{sw}_\Gamma\big(K^\top x \oplus S^\top y\big) \equiv 4x_1x_2x_3 \pmod 8.$$

requirement	reason
phase independent of y	all terms of one coset sum must keep their relative phase
phase equal to $(-1)^{x_1x_2x_3}$	the logical action must be CCZ

XOR turns the phase into overlap counts

For rows a, b, the coordinatewise product $a \wedge b$ marks common ones. At each coordinate, as integers,

$$a \oplus b = a + b - 2\,a \wedge b, \qquad a \oplus b \oplus c = a + b + c - 2(a \wedge b + a \wedge c + b \wedge c) + 4\,a \wedge b \wedge c.$$

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In general, an XOR of m rows expands with coefficient $(-2)^{t-1}$ on each t-fold product. The signed weight sw_Γ is a weighted sum of coordinates, so it respects these integer identities:

$$\mathrm{sw}_\Gamma(a \oplus b \oplus c) = \textstyle\sum_{\text{rows}} \mathrm{sw}_\Gamma(a) - 2 \sum_{\text{pairs}} \mathrm{sw}_\Gamma(a \wedge b) + 4\ \mathrm{sw}_\Gamma(a \wedge b \wedge c).$$

Products of four or more rows have coefficients divisible by 8 and drop out mod 8.

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Products of four or more rows have coefficients divisible by 8 and drop out mod 8.

Check: supports $\{1,2\}, \{2,3\}, \{2,4\}$ have weights 2, pair overlaps 1, triple overlap 1: $|a \oplus b \oplus c| = 6 - 2(3) + 4 = 4$.

Signed overlaps decide the native gate

A codeword $K^\top x \oplus S^\top y$ is the XOR of the rows g selected by (x, y). Matching coefficients of 1, -2, and 4 against $4x_1x_2x_3$ gives:

Native criterion (paper, Prop. signed overlaps)

Γ implements CCZ natively if and only if, for distinct rows g, g', g'' of G,

$$\mathrm{sw}_\Gamma(g) \equiv 0 \pmod 8,$$

$$\mathrm{sw}_\Gamma(g \wedge g') \equiv 0 \pmod 4,$$

$$\mathrm{sw}_\Gamma(g \wedge g' \wedge g'') \equiv \begin{cases} 1 \pmod 2, & \{g, g', g''\} = \{K_1, K_2, K_3\}, \\ 0 \pmod 2, & \text{otherwise.} \end{cases}$$

In the cube code, $\Gamma = (-1)^p$ meets all three, with the logical triple $u_1 \wedge u_2 \wedge u_3$ at the single site 111.

Quasi-transversal versus native

Quasi-transversal: apply $T^{\otimes n}$, then a diagonal Clifford correction built from $P = \mathrm{diag}(1, i)$ and CZ gates. Campbell and Howard give nine parity conditions on G that suffice. Hu, Liang, and Calderbank show they are also necessary:

logical rows	mixed	stabilizer rows
K1: $\lvert K_i \rvert$ even	KS1: $\lvert K_i \wedge S_l \rvert$ even	S1: $\lvert S_l \rvert$ even
K2: $\lvert K_i \wedge K_{i'} \rvert$ even	KS2: $\lvert K_i \wedge K_{i'} \wedge S_l \rvert$ even	S2: $\lvert S_l \wedge S_{l'} \rvert$ even
K3: $\lvert K_1 \wedge K_2 \wedge K_3 \rvert$ **odd**	KS3: $\lvert K_i \wedge S_l \wedge S_{l'} \rvert$ even	S3: $\lvert S_l \wedge S_{l'} \wedge S_{l''} \rvert$ even

Native is the special case whose correction is $P^\dagger$ on single qubits, since $TP^\dagger = T^\dagger$. So every native code meets the nine conditions. The converse fails (slide 19).

Index ranges?

One Boolean function packages every overlap

Let G have r rows and distinct columns. Its *column indicator* $\chi_G\colon \mathbb{F}_2^r \to \mathbb{F}_2$ equals 1 exactly on the columns of G. For a test polynomial m in the row variables $z_1, \dots, z_r$,

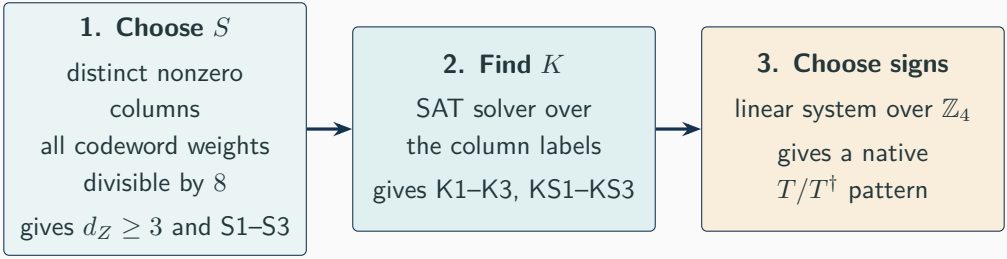
$$\langle \chi_G, m \rangle = \sum_{\text{columns } z} m(z) \pmod 2.$$

test polynomial m	$\langle \chi_G, m \rangle$ is the parity of
z_i	the weight of row i
$z_i z_j$	the overlap of rows i and j
$z_i z_j z_l$	the triple overlap of rows i, j, l

The nine conditions prescribe $\langle \chi_G, m \rangle$ for every monomial of degree 1 to 3.

A code becomes a set of points in $\mathbb{F}_2^r$. Its overlap conditions become degree conditions on one Boolean function (Nezami–Haah).

The construction runs in three steps



Each step fixes one group of conditions and never revisits the previous one.

Geometry settles the distance before any search begins.

Distinct nonzero columns give $d_Z \geq 3$

Column j of S is the syndrome $\beta_j \in \mathbb{F}_2^s$ of a Z error on qubit j. A Z error v of weight at most two has syndrome

$$Sv = \beta_j \qquad \text{or} \qquad Sv = \beta_i \oplus \beta_j.$$

If the columns are distinct and nonzero, both are nonzero: every such error is detected, whatever K is.

Such a column set is a set of points of the *projective space* $\mathrm{PG}(s-1, 2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$. The code $\mathcal{C}_2$ is then called *projective*.

Cube code: $s = 1$ and all eight columns equal 1. The pair $\{000, 100\}$ has syndrome $1 \oplus 1 = 0$, which is the distance-2 error of slide 4.

Projective S gives $d_Z \geq 3$ for every choice of logical rows.

Weights are read from hyperplane sections

Call $\mathcal{C}_2$ *8-divisible* if every codeword weight is a multiple of 8. Then S1–S3 hold:

$$|a| \equiv 0 \ (8) \ \Rightarrow \ |a \wedge b| \equiv 0 \ (4) \ \Rightarrow \ |a \wedge b \wedge c| \equiv 0 \ (2),$$

using the pair and triple identities of slide 7 on $|a \oplus b|$ and $|a \oplus b \oplus c|$.

Choose a point set X whose hyperplane sections all have sizes $\equiv |X| \pmod 8$.

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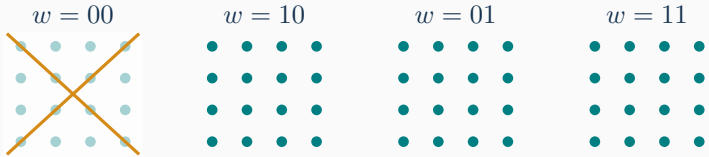
A nonzero $a \in \mathbb{F}_2^s$ gives the codeword $c_a = a^\top S$ with entries $c_a(j) = a \cdot \beta_j$. Let X be the column set and $H_a = \{\beta \neq 0 : a \cdot \beta = 0\}$, a hyperplane. Then

$$|c_a| = |X| - |X \cap H_a|.$$

Choose a point set X whose hyperplane sections all have sizes $\equiv |X| \pmod 8$.

FG48: projective space minus a solid

Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. Remove the 15 nonzero points with $w = 00$, a solid $\mathrm{PG}(3, 2)$:

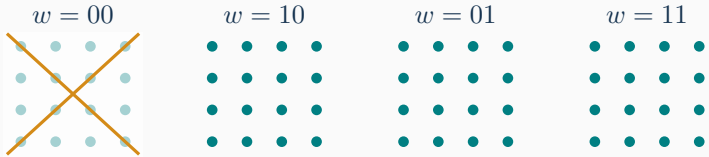


$63 - 15 = 48$ columns, $s = 6$. The rows of S are the coordinate functions $u_1, \ldots, u_4, w_1, w_2$.

Why only weights 24 and 32?

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A linear form $\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w$ has weight

$$\lambda \neq 0 : \ 3 \cdot 8 = 24, \qquad \lambda = 0, \ \mu \neq 0 : \ 2 \cdot 16 = 32.$$

FG48 is projective and 8-divisible. Up to coordinate permutation it is the only projective 8-divisible $[48, 6]$ code.

Why only weights 24 and 32?

Logical rows by SAT

Unknowns: a logical label $\alpha_j \in \mathbb{F}_2^3$ for each column j, the jth column of K.

Already settled by S

- $d_Z \geq 3$ for every labeling
- S1–S3 constrain no label

Left for the solver

- K1, K2, K3: 7 parities
- KS1, KS2: $6s$ parities
- KS3: $3\binom{s}{2}$ parities

Each condition is a parity of products of label bits, of degree at most three. For FG48 ($n = 48$, $s = 6$):

$3n = 144$ Boolean variables, $\quad 7 + 6s + 3\binom{s}{2} = 88$ constraints, no distance constraint.

Fixing a projective, 8-divisible S first turns the code search into a small satisfiability problem.

The logical rows of Q_{48}

On each block, the three logical rows are Boolean polynomials in $u = (u_1, u_2, u_3, u_4)$:

	$w = 10$	$w = 01$	$w = 11$
K_1	$u_1 \oplus u_3$	$1 \oplus u_3$	u_4
K_2	$1 \oplus u_1 \oplus u_2 \oplus u_3 \oplus u_4$	$1 \oplus u_1 \oplus u_3 \oplus u_1u_3 \oplus u_1u_4$	$u_3 \oplus u_4 \oplus u_1u_3 \oplus u_1u_4$
K_3	$1 \oplus u_1 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4$	$u_1 \oplus u_1u_2 \oplus u_2u_3 \oplus u_2u_4 \oplus u_3u_4$	$u_1 \oplus u_3 \oplus u_1u_4 \oplus u_2u_3 \oplus u_2u_4$

Degree at most two on each block. The eight labels x give eight disjoint cosets $K^\top x \oplus \mathcal{C}_2$.

$K_1 \wedge K_2 \wedge K_3$ has odd weight (K3). Every other overlap parity is even.

Signs from a linear system over $\mathbb{Z}_4$

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$ marking $T^\dagger$. Then $\mathrm{sw}_\Gamma(c) = |c| - 2\,\gamma \cdot c$, and native CCZ becomes

$$\gamma \cdot c \equiv \tfrac{1}{2}|c| - 2x_1x_2x_3 \pmod 4 \qquad \text{for every codeword } c = K^\top x \oplus S^\top y.$$

For Q_{48}, a solution is $\gamma(u, w) = u_3 \oplus w_1 Q_1(u) \oplus w_2 Q_2(u)$ with

$$Q_1 = u_1 \oplus u_2 \oplus u_3 \oplus u_1u_3 \oplus u_2u_3 \oplus u_2u_4, \quad Q_2 = u_3 \oplus u_4 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4.$$

$w = (1,0)$

T	T	T	T
$T^\dagger$	T	$T^\dagger$	T
T	$T^\dagger$	$T^\dagger$	T
$T^\dagger$	$T^\dagger$	T	T

6 $T^\dagger$

$w = (0,1)$

T	$T^\dagger$	T	T
T	$T^\dagger$	T	T
$T^\dagger$	$T^\dagger$	T	$T^\dagger$
T	T	$T^\dagger$	T

6 $T^\dagger$

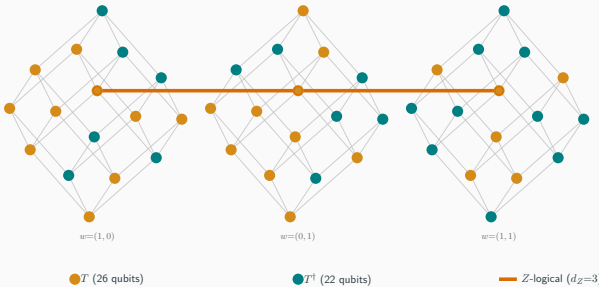
$w = (1,1)$

T	$T^\dagger$	$T^\dagger$	$T^\dagger$
$T^\dagger$	$T^\dagger$	T	$T^\dagger$
$T^\dagger$	T	T	T
$T^\dagger$	$T^\dagger$	T	$T^\dagger$

10 $T^\dagger$

Gold: T. Teal: $T^\dagger$. Rows u_1u_2, columns u_3u_4, both ordered $00, 01, 11, 10$.

The code $Q_{48} = [\![48, 3, 3]\!]$



Three 4-cubes (blocks $w = 10, 01, 11$). Gold: T. Teal: $T^\dagger$.

$d_X = 16$, $d_Z = 3$, native CCZ by $26\,T + 22\,T^\dagger$.

A weight-3 Z logical (orange): qubits $u = 0000$ in the three blocks.
Syndromes: w parts $10 \oplus 01 \oplus 11 = 00$, u parts zero.
Labels: $011 \oplus 110 \oplus 000 = 101 \neq 0$.
Projectivity gives $d_Z \geq 3$, so $d_Z = 3$.

Checked directly: all 8×64 codewords satisfy the phase condition.

The nine conditions do not imply native

Jacinto et al. give a $[\![47, 3, 3]\!]$ code with $d_X = 16$, $d_Z = 3$, $s = 6$, and projective support. It meets all nine conditions.

For it, the $\mathbb{Z}_4$ system of slide 17 has 47 unknowns and one equation per codeword. Elimination over $\mathbb{Z}_4$ shows it is inconsistent.

- No $T/T^\dagger$ pattern implements CCZ on this code.
- Any diagonal Clifford correction for $T^{\otimes 47}$ needs entangling CZ gates.

The smallest known code meeting the nine conditions has 47 qubits. The smallest known native code has 48.

Stabilizer conditions restrict the length

S1–S3 say: $\langle \chi_S, m \rangle = 0$ for every monomial m of degree 1 to 3. The dual of the Reed–Muller code $\mathrm{RM}(3, s)$ is $\mathrm{RM}(s-4, s)$. With δ_0 the indicator of the origin:

Admissible lengths (paper, Thm.)

S1–S3 hold iff $\chi_S \in \mathrm{RM}(s-4, s) + \langle \delta_0 \rangle$. The lengths are $n = |c| - c(0)$ for $c \in \mathrm{RM}(s-4, s)$ whose support spans $\mathbb{F}_2^s$.

s	code	weights ≤ 39	candidate $15 \leq n \leq 38$
5	$\mathrm{RM}(1,5)$	$16, 32$	$15, 16, 31$
6	$\mathrm{RM}(2,6)$	$16, 24, 28, 32, 36$	$15, 16, 23, 24, 27, 28, 31, 32, 35, 36$
7	$\mathrm{RM}(3,7)$	$16, 24, 28, 32, 36$	$15, 16, 23, 24, 27, 28, 31, 32, 35, 36$

No code below length 39

Theorem (length lower bound)

No CSS code with $k = 3$ and $d_Z \geq 3$ meets the nine conditions at $n \leq 38$, at any stabilizer rank s.

1. Make $\mathcal{C}_2$ projective and lower s while $2^s - 1 > n + \lfloor \binom{n}{2}/2 \rfloor$. At $n \leq 38$ this removes every $s \geq 9$.
2. $s \leq 5$: few admissible supports, all flats but one, and a direct search settles the last.
3. Supports that are affine flats of dimension at most 4 fail K3. This removes $n = 15, 16$.
4. $s = 6, 7, 8$: one support per affine-equivalence class of $c \in \mathrm{RM}(s-4, s)$, then an exact computer-assisted label search on each.

The bound covers quasi-transversal codes, so it covers native codes too.

Why one support per class?

A family with all-T CCZ

Generalize FG48: delete a p-dimensional subspace from $\mathbb{F}_2^s$, $s = p + q$. The columns are $(u, w) \in \mathbb{F}_2^p \times \mathbb{F}_2^q$ with $w \neq 0$, and S has the coordinate rows.

$$K_1 = w_1 u_1 u_2, \qquad K_2 = w_1 u_3 u_4, \qquad K_3 = w_1 \Big(\prod_{i=5}^{p} u_i\Big)\Big(\prod_{j=2}^{q} w_j\Big).$$

Proposition (paper)

For $p \geq 4$, $q \geq 1$, $s \geq 7$: the nine conditions hold, $d_Z \geq 3$, and T **on every qubit** implements CCZ.

Length $n = 2^s - 2^p$. At $s = 7$: $n = 64, 96, 112$. The smallest member has $n = 64$.

Q_{48} has $s = 6$, below this family, and needs 22 $T^\dagger$ gates.

Summary and open problems

Results

- $Q_{48} = [\![48, 3, 3]\!]$ with native CCZ by $26\,T + 22\,T^\dagger$, the shortest known native code.
- No code meets the nine conditions with $k = 3$, $d_Z \geq 3$, $n \leq 38$.
- The $[\![47, 3, 3]\!]$ code meets them but has no native pattern.

Open

- Lengths $39 \leq n \leq 46$.
- Which subspace complements with $n \geq 48$ admit native CCZ, and the largest d_X on a fixed support.
- Other non-Clifford gates, and more than three logical qubits.

Boolean functions turn the gate into overlap counts. Projective geometry makes those counts controllable.

Where can I read more?

The nine conditions with index ranges

Question from slide 9: which rows enter each condition?

label	condition
K1	$\lvert K_i \rvert \equiv 0 \pmod 2$
K2	$\lvert K_i \wedge K_{i'} \rvert \equiv 0 \pmod 2$
K3	$\lvert K_1 \wedge K_2 \wedge K_3 \rvert \equiv 1 \pmod 2$
KS1	$\lvert K_i \wedge S_l \rvert \equiv 0 \pmod 2$
KS2	$\lvert K_i \wedge K_{i'} \wedge S_l \rvert \equiv 0 \pmod 2$
KS3	$\lvert K_i \wedge S_l \wedge S_{l'} \rvert \equiv 0 \pmod 2$
S1–S3	$\lvert S_l \rvert$, $\lvert S_l \wedge S_{l'} \rvert$, $\lvert S_l \wedge S_{l'} \wedge S_{l''} \rvert \equiv 0 \pmod 2$

Indices: $1 \le i < i' \le 3$ and $1 \le l < l' < l'' \le s$. K3 is the single odd parity.

▸ Return to slide 9

Why FG48 has only weights 24 and 32

Question from slide 14: why only two weights, and why is FG48 unique?

The 48 columns occupy 48 of the 63 points of $\mathrm{PG}(5,2)$, missing 15. Each hyperplane has 31 points, so it meets the column set in at least $31-15=16$ and at most 31 points. 8-divisibility forces the section size $48-|c_a|$ to be a multiple of 8: 16 or 24, giving weights 32 or 24.

Count incidences: 63 hyperplanes, each of the 48 points on 31 of them:

$$N_{16}+N_{24}=63, \qquad 16N_{16}+24N_{24}=48\cdot 31 \quad\Longrightarrow\quad N_{16}=3,\ N_{24}=60.$$

Every projective 8-divisible $[48,6]$ code has this weight enumerator. The classification of projective 8-divisible codes of length at most 48 leaves only the solid complement.

Calderbank–Kantor 1986 Betsumiya–Munemasa 2012 Kurz 2021.

▸ Return to slide 14

Backup

Affine changes of coordinates preserve everything

Question from slide 21: why is one support per affine-equivalence class enough?

Let A be an invertible binary $s \times s$ matrix and $t \in \mathbb{F}_2^s$, with arithmetic mod 2. The map $\beta \mapsto A\beta \oplus t$ sends the support of c to the support of $c'(\beta) = c(A\beta \oplus t)$.

- It permutes points, so it preserves weights and the Reed–Muller degree.
- A linear change A replaces S by AS up to column order. AS has the same row space $\mathcal{C}_2$.
- A translation t moves the origin. The paper therefore checks each translate $\mathcal{P}_a$ of a representative and its actual rank.

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Sources

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